Chapter 4 exercises

DS04-01 • The last queue node

A linked queue stores head and tail. From empty, enqueue 4, dequeue, then enqueue 9. Draw pointers and ownership at every state. Explain the failure if removing the last node updates only head.

DS04-02 • Proving in-place reversal

Reverse `3→8→11→null` using three pointers. List the lists reached by prev and curr after every iteration. State the disjointness and coverage invariant. How must tail change if it is also stored?

DS04-03 • Polynomial normalization and addition

P has input terms `(4,2),(1,1),(4,-1)`. Q has `(4,-1),(0,5)`. Explain why P is not canonical. Find canonical P+Q, and analyze sorting and merging separately.

DS04-04 • Doubly linked deletion

A doubly linked list has head and tail sentinels, and x is a known data node. Give O(1) deletion steps and prove consistency of the remaining neighbors. State invalid x or sentinel states that the API contract must exclude.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

References and attribution

Ellis Horowitz, Sartaj Sahni, Dinesh Mehta. Fundamentals of Data Structures in C++, 2nd Edition.

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